ASSIGNMENT 6

Implementation of Data Integrity Constraints

HOSPITAL MANAGEMENT SYSTEM

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Course: BCA (Hons) | Subject: Database Management System

1. Objective

Integrity constraints keep invalid values out of the database and prevent records from referring to missing parent records. The following constraints are implemented in the Assignment 5 schema.

2. Constraint inventory

Constraint	Where used	Purpose
PRIMARY KEY	All eight tables	Uniquely identifies each row and disallows NULL key values.
FOREIGN KEY	Doctors, Appointments, Prescriptions, Bills, Feedbacks	Ensures each reference points to an existing parent row.
UNIQUE	Departments.department_name; Doctors.email; Prescriptions(appointment_id, medicine_id)	Prevents duplicate department names, email addresses and repeated medicine lines per appointment.
CHECK	Appointment status	Allows only Scheduled, Completed, Cancelled or No-show.
CHECK	Medicines.price; Bills.total_amount	Prevents negative prices and bill totals.
CHECK	Prescriptions.quantity	Requires a positive quantity.
CHECK	Feedbacks.rating	Restricts rating to the range 1–5.
NOT NULL / DEFAULT	Required fields and status fields	Requires essential values and supplies defaults where defined.

3. SQL statements to verify constraints

```
USE hospital_management;

-- Inspect table definitions and constraints
SHOW CREATE TABLE Doctors;
SHOW CREATE TABLE Appointments;
SHOW CREATE TABLE Prescriptions;
SHOW CREATE TABLE Bills;
SHOW CREATE TABLE Feedbacks;

-- Verify no orphan appointments exist
```



```
SELECT a.appointment_id
FROM Appointments a
LEFT JOIN Patients p ON p.patient_id = a.patient_id
LEFT JOIN Doctors d ON d.doctor_id = a.doctor_id
WHERE p.patient_id IS NULL OR d.doctor_id IS NULL;
```

```
-- Verify ratings and quantities in stored data
SELECT * FROM Feedbacks WHERE rating < 1 OR rating > 5;
SELECT * FROM Prescriptions WHERE quantity <= 0;
SELECT * FROM Medicines WHERE price < 0;
SELECT * FROM Bills WHERE total_amount < 0;
```

4. Controlled negative tests

Run these one at a time only in a test database. Each should fail because it violates a constraint; do not use them as production inserts.

```
-- Unknown department: foreign-key violation
INSERT INTO Doctors (doctor_id, department_id, name, specialization, phone, email)
VALUES (999, 999, 'Test Doctor', 'General', '9000000000', 'test@example.com');

-- Invalid rating: CHECK violation
INSERT INTO Feedbacks (feedback_id, patient_id, doctor_id, rating, comments)
VALUES (999, 1, 1, 6, 'Invalid test');

-- Negative medicine price: CHECK violation
INSERT INTO Medicines (medicine_id, medicine_name, manufacturer, price)
VALUES (999, 'Test Medicine', 'Test Co', -1.00);
```

5. Expected results and explanation

The orphan-record query and the invalid-value queries should return zero rows when data is valid. SHOW CREATE TABLE should show the defined keys and constraints. The negative test inserts should be rejected by MySQL. Constraint error text can vary by server version and client.

Conclusion

Primary keys, foreign keys, unique rules, checks and required columns collectively protect entity identity, relationships and domain validity. Constraints should be tested on a non-production database before deployment.

ASSIGNMENT 6 — OUTPUTS AND VERIFICATION

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Output 1 — Orphan appointments

Query result: appointment_id
(0 rows)

Output 2 — Invalid stored values

Validation query	Expected result
Feedbacks WHERE rating < 1 OR rating > 5	0 rows
Prescriptions WHERE quantity <= 0	0 rows
Medicines WHERE price < 0	0 rows
Bills WHERE total_amount < 0	0 rows

Output 3 — Controlled negative tests

Test	Expected MySQL result
Insert doctor with department_id = 999	Rejected: foreign-key constraint violation
Insert feedback rating = 6	Rejected: CHECK constraint violation
Insert medicine price = -1.00	Rejected: CHECK constraint violation

Conclusion

Result: valid sample data passes the verification queries with zero invalid rows. Invalid test inserts should be rejected by the declared constraints. Exact error wording depends on MySQL version.